

Representative-based Clustering

Given a dataset with n points in a d -dimensional space, $\mathbf{D} = \{\mathbf{x}_i\}_{i=1}^n$, and given the number of desired clusters k , the goal of representative-based clustering is to partition the dataset into k groups or clusters, which is called a *clustering* and is denoted as $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$.

For each cluster C_i there exists a representative point that summarizes the cluster, a common choice being the mean (also called the *centroid*) μ_i of all points in the cluster, that is,

$$\mu_i = \frac{1}{n_i} \sum_{\mathbf{x}_j \in C_i} \mathbf{x}_j$$

where $n_i = |C_i|$ is the number of points in cluster C_i .

A brute-force or exhaustive algorithm for finding a good clustering is simply to generate all possible partitions of n points into k clusters, evaluate some optimization score for each of them, and retain the clustering that yields the best score. However, this is clearly infeasible, since there are $O(k^n/k!)$ clusterings of n points into k groups.

A brute-force or exhaustive algorithm for finding a good clustering is simply to generate all possible partitions of n points into k clusters, evaluate some optimization score for each of them, and retain the clustering that yields the best score. The *exact* number of ways of partitioning n points into k nonempty and disjoint parts is given by the *Stirling numbers of the second kind*, given as

$$S(n, k) = \frac{1}{k!} \sum_{t=0}^k (-1)^t \binom{k}{t} (k-t)^n$$

For $k = 2$, the formula gives

$$S(n, 2) = \frac{1}{2} (2^n - 2 \cdot 1^n) = 2^{n-1} - 1.$$

For $n = 4$ this is $2^3 - 1 = 7$ (matches the earlier explicit list). For $n = 10$ this is $2^9 - 1 = 511$.

For small fixed k the approximation $S(n, k) \approx \frac{k^n}{k!}$ becomes very good as n grows.

K-means Algorithm: Objective

The *sum of squared errors* scoring function is defined as

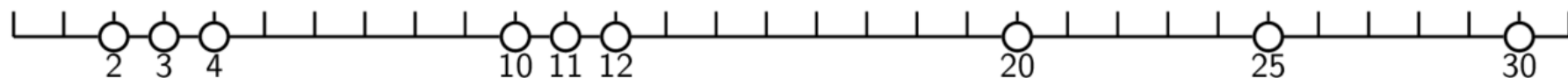
$$SSE(\mathcal{C}) = \sum_{i=1}^k \sum_{\mathbf{x}_j \in C_i} \|\mathbf{x}_j - \boldsymbol{\mu}_i\|^2$$

The goal is to find the clustering that minimizes the SSE score:

$$\mathcal{C}^* = \arg \min_{\mathcal{C}} \{SSE(\mathcal{C})\}$$

K-means employs a greedy iterative approach to find a clustering that minimizes the SSE objective. As such it can converge to a local optima instead of a globally optimal clustering.

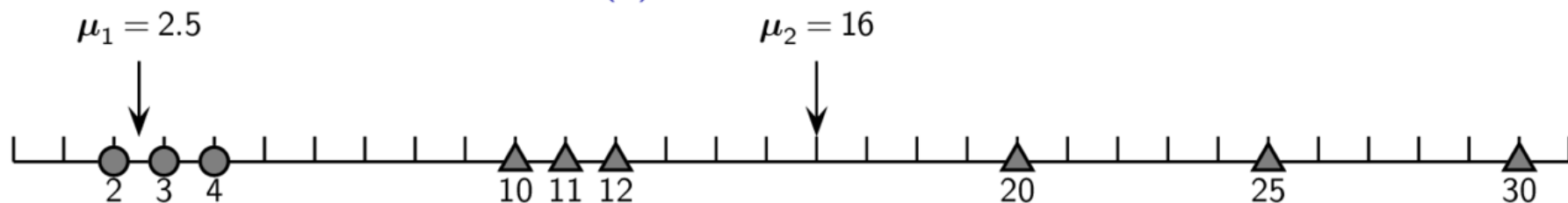
K-means in One Dimension



(a) Initial dataset

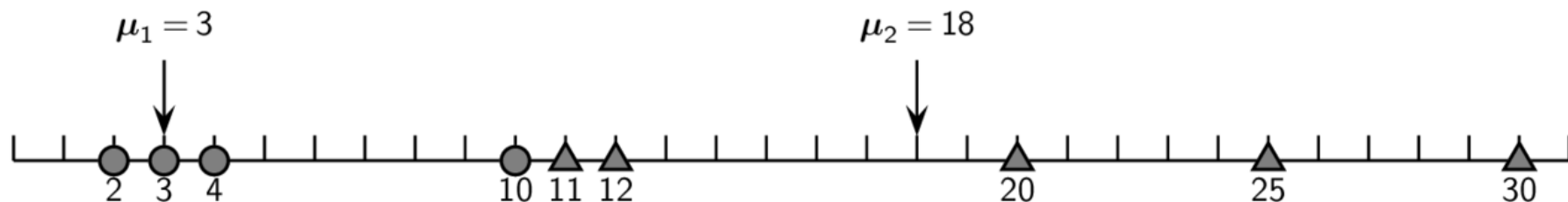


(b) Iteration: $t = 1$



(c) Iteration: $t = 2$

K-means in One Dimension (contd.)



(d) Iteration: $t = 3$



(e) Iteration: $t = 4$



(f) Iteration: $t = 5$ (converged)

K-means Algorithm: Objective

K-means initializes the cluster means by randomly generating k points in the data space. Each iteration of K-means consists of two steps: (1) cluster assignment, and (2) centroid update.

Given the k cluster means, in the cluster assignment step, each point $\mathbf{x}_j \in \mathbf{D}$ is assigned to the closest mean, which induces a clustering, with each cluster C_i comprising points that are closer to μ_i than any other cluster mean. That is, each point $\mathbf{x}_j$ is assigned to cluster C_{i^*} , where

$$i^* = \arg \min_{i=1}^k \left\{ \|\mathbf{x}_j - \mu_i\|^2 \right\}$$

Given a set of clusters C_i , $i = 1, \dots, k$, in the centroid update step, new mean values are computed for each cluster from the points in C_i .

The cluster assignment and centroid update steps are carried out iteratively until we reach a fixed point or local minima.

K-Means Algorithm

K-means (D, k, ϵ):

```
1  $t = 0$ 
2 Randomly initialize  $k$  centroids:  $\mu_1^t, \mu_2^t, \dots, \mu_k^t \in \mathbb{R}^d$ 
3 repeat
4    $t \leftarrow t + 1$ 
5    $C_j \leftarrow \emptyset$  for all  $j = 1, \dots, k$ 
   // Cluster Assignment Step
6   foreach  $\mathbf{x}_j \in D$  do
7      $i^* \leftarrow \operatorname{argmin}_i \left\{ \|\mathbf{x}_j - \mu_i^t\|^2 \right\}$  // Assign  $\mathbf{x}_j$  to closest
       centroid
8      $C_{i^*} \leftarrow C_{i^*} \cup \{\mathbf{x}_j\}$ 
   // Centroid Update Step
9   foreach  $i = 1$  to  $k$  do
10     $\mu_i^t \leftarrow \frac{1}{|C_i|} \sum_{\mathbf{x}_j \in C_i} \mathbf{x}_j$ 
11 until  $\sum_{i=1}^k \|\mu_i^t - \mu_i^{t-1}\|^2 \leq \epsilon$ 
```

Jai Sairam